

Rope X: Assistance and Guidance on Jumping Rope Frequency, based on Real-time, Heart Rate Feedback During Exercise

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ABSTRACT

Jumping rope is a very efficient aerobic exercise, during which the exerciser can maintain an efficient heart rate range (heart rate value to 140 ~ 160bpm). However, at present, people are unable to recognize whether they are in this ideal aerobic state in a timely and accurate manner, and there is no convenient way of showing exercisers how to perform efficient jumping rope exercises. We developed an early prototype, Rope X, an exercise device that communicates the exerciser's heart rate status through a dynamic interactive interface and gives the exerciser a certain degree of guidance, as to the correct use of the jumping rope to improve the efficiency of the jumping rope exercise (aerobic exercise).

CCS CONCEPTS

• Human-centred computing; • Human computer interaction (HCI); • Interaction devices;

KEYWORDS

Smart Exercise Interactive Device, Aerobic Exercise, Smart Jump Rope, Real Time Heart Rate Feedback, Interface Free Design

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1 INTRODUCTION

In recent years, there are a number of studies that use heart rate testing to guide people to moderate amounts of aerobic exercise [6, 8]. Jumping rope as a popular sport contains both anaerobic and aerobic states [1] [3] [4]. Aerobic state of jumping rope can only be achieved efficiently if the exerciser maintains a certain exercise duration and keeps the heart rate stable in the 140-160 bpm range during the process [7]. At present, the heart rate data obtained by most exercisers, using external devices such as smart watches or heart rate bands, tend to lag and cannot effectively guide the exercise process.

2 DESIGN/IMPLEMENT

2.1 System

The prototype, Rope X, collects the heart rate data of the exerciser during the jumping rope process in real-time. The heart rate of the exerciser in different heart rate zones, will be converted into different colours and displayed on the rope; a blue colour is used for a heart rate of 60~100bpm, yellow for 100~140bpm, green for 140~160bpm (ideal) and red for a heart rate above 160bpm (Figure 1). Rope X uses visual feedback relating to the colour to inform the exerciser of their current heart rate.

We simultaneously use vibrations to guide the user's frequency of movement. Based on the study of the relationship between jumping rope frequency and aerobic exercise heart rate [2] [5], When jumping rope at a frequency of 120 bpm, the movement in the heart

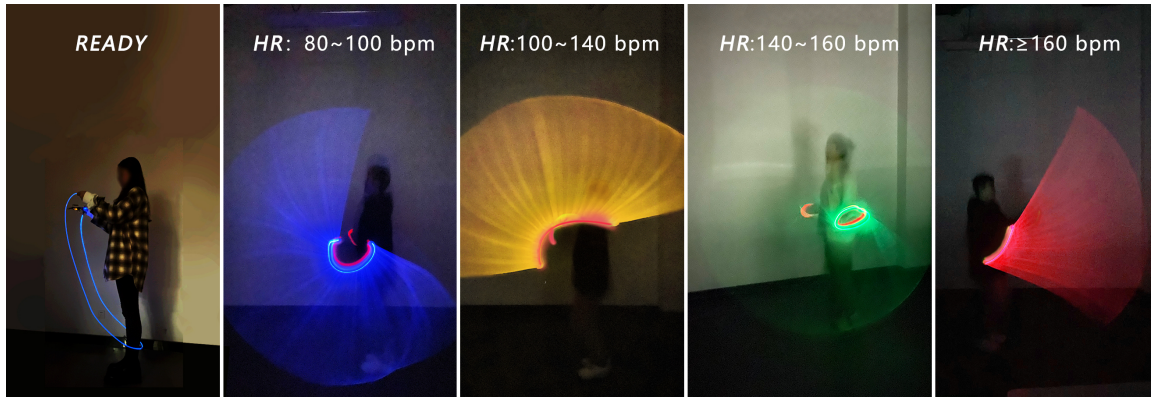


Figure 1: The colour change of the jumping rope depending on the heart rate. Blue (HR: 80~100 bpm); yellow (HR: 100~140 bpm); green (HR: 140~160 bpm); red (HR: ≥ 160 bpm).

Table 1: Heart rate change and vibration feedback

Heart Rate(HR)	Tip colour	Handle vibration frequency	Feedback to regulate exercise
60~100 bpm	Blue	135 bpm	Low - Change the vibration frequency to increase the HR
100~140 bpm	Yellow	135 bpm	
140~160 bpm	Green	120 bpm	Perfect - Keep vibrating to stabilize the HR
≥ 160 bpm	Red	80 bpm	High - Change the vibration frequency to moderate the HR

^a This is a table of the colours and vibration frequencies, corresponding to real-time, heart rate zones.

rate can be stable and in the ideal range of 140 ~ 160 bpm. We set the frequency parameters of vibration feedback on this basis.

Table 1 shows the corresponding relationship between different heart rate zones, rope colour and vibration frequency during exercise. We aim to help the exercisers change the rhythm of jumping rope, by increasing or decreasing the vibration frequency, assisting exercisers in stabilizing their heart rate within an ideal range.

3 PROTOTYPE

Rope X was our early prototype in this study. We added a blood oxygen heart rate sensor and a vibration motor to the two handles respectively, and then connected the LED light, transmitting the data from the sensors in the left and right handles, by means of a Bluetooth connection. At the same time, we replaced the rope with optical fibre, for achieving the effect of better colour display of the rope. Rope X emulates regular jumping rope. The handle is held with both hands and one thumb is placed on the heart rate sensing module.

The electronic equipment hardware of the prototype includes two Arduino Nanos, a MAX30102 blood oxygen heart rate sensor, a vibration motor, an RGB LED lighting module, a set of HC-05 Bluetooth modules and two 9V batteries. The logic of using Rope X and the hardware of the electronic devices is shown in Figure 2. The prototype is housed within a handle measuring 34 x 34 x 156mm in size, with a rope-fibre section, approximately 3m in length.

4 SYSTEM FEASIBILITY TESTING

To investigate whether our prototypes provide clear and effective guidance to exercisers, we conducted a few simple tests. We found that with rope X guidance, exercisers were more likely to reach an aerobic heart rate state and were able to stay in the aerobic heart rate zone for relatively long periods of time.

The positive feedback, received from the participants, was encouraging. The experiment and interview results show that the multi-sensory interaction, using real-time, heart rate feedback to improve and guide exercisers to maintain a perfect heart rate, was positive. The three subjects also indicated that the prototype's guidance of the exercise was both natural and obvious and that they are likely to choose Rope X, which is an exciting comment. This shows that Rope X has a considerable effect on improving the efficiency of aerobic exercise and is particularly significant for our later, in-depth study

5 FUTURE WORK AND CONCLUSION

In summary, our preliminary study presents several promising and efficient results. Obtaining timely and accurate real-time, heart rate feedback during exercise, has a beneficial effect and using real-time, heart rate feedback during exercise to provide interaction and guidance, can effectively improve aerobic efficiency. These preliminary results constitute a contribution to the field of smart sport interaction,

Our researchers also discovered certain equipment problems during the prototyping and testing process, for example, the light transmission of the optical fibre was affected by the environment and the

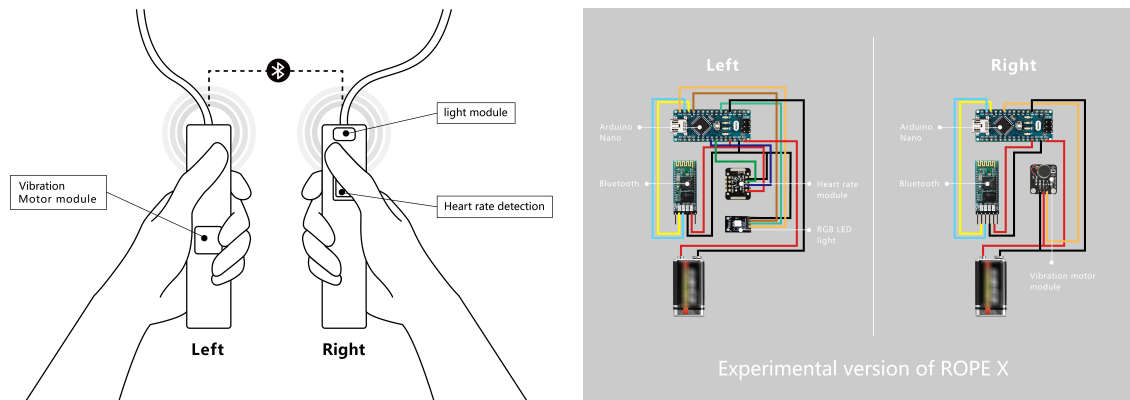


Figure 2: The internal structure of the handle is matched with electronic hardware. (Experimental version)

participants could not effectively identify the colour changes; there were some inaccuracies in the heart rate data obtained by Rope X. Although this did not affect our experimental results, further refinement was required in later iterations of the prototype.

In future work, as well as improving current problems, we will conduct more in-depth research and iterations to explore the usability and possibilities of interaction and figure out feedback in the field of sports and health. At the same time, based on research into the jumping rope exercise, we explored other aerobic sports in the form of intelligent sports interaction research and identified the implementation of real-time, biological data feedback in the field of aerobic sports and intelligent sports interaction equipment as an area for more detailed research. We also hope that intelligent sports interaction equipment and interface-free interaction will be more effective in the future.

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